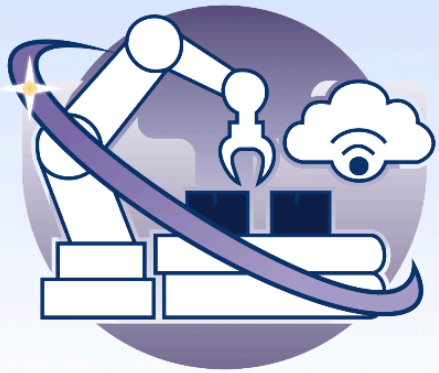
DEPARTEMEN TEKNIK ELEKTRO OTOMASI

GUIDEBOOK COMPETITION IARC XII



AUTOMATION CHALLENGE

SD SEDERAJAT
SMP SEDERAJAT

SMA/K SEDERAJAT
MAHASISWA

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COMPETITION OVERVIEW

The Internet of Things (IoT) is a concept that connects various physical devices through the internet, enabling processes such as data acquisition, communication, monitoring, controlling, as well as automated and real-time decision-making.

The application of this concept is one of the approaches used for idea development in the **Automation Challenge Advance – Microcontroller Competition**, based on the research focuses of three Pusat Studi (Pusdi) under Direktorat Riset dan Pengabdian kepada Masyarakat (DRPM) Sepuluh Nopember Institute of Technology, which are:

- Energy and Mineral Resources
- Disaster Mitigation and Climate Change
- Appropriate Technology

These three fields serve as the basis for determining problems, ensuring that the proposed solutions are not only technologically superior, but also innovative, efficient, applicable, and capable of providing benefits to society and the environment.

The **Automation Challenge Advance – Microcontroller Competition IARC XII** is a technology innovation competition focused on the participants' ability to design, develop, and directly implement Internet of Things (IoT)-based solutions. Unlike conventional paper-writing competitions, this event requires participants to physically assemble and test their designed devices in a real setting.

This competition evaluates not only the aspects of innovation and proposal alignment, but also the TRL (Technology Readiness Level / *Tingkat Kesiapan Teknologi*), such as the precision of component selection, implementation feasibility, successful device testing, and the impact of the created solution on society.

EVENT OBJECTIVES

The objectives of this event are:

1. To encourage participants to develop well-integrated, microcontroller-based automation systems that can be implemented in the real world.
2. To train participants' skills in analyzing and designing interconnective microcontroller automation architectures, ranging from sensor data acquisition and actuator control to smart monitoring systems.
3. To serve as a platform for students to hone their creativity and technical skills in the fields of technology and industrial automation.
4. To provide hands-on experience in designing, building, and presenting microcontroller prototypes directly in front of the judges and visitors.
5. To build a competitive, collaborative, and innovative academic atmosphere in the field of microcontroller technology.
6. To serve as a platform for developing creative ideas relevant to the theme *"Innovative Microcontroller for Sustainable Living: Energy, Disaster Mitigation, and Appropriate Technology."*

XII IARC Industrial Automation and Robotic Competition

EVENT TIMELINE

Event	Timeline/Date
Open Registration Batch 1	September 25 – October 16, 2026
Open Registration Batch 2	October 17 – November 7, 2026
Open Registration Batch 3	November 8 – November 27, 2026
Proposal Submission	December 11, 2026
Top 10 Finalists Announcemen	December 16, 2026
Technical Meeting Final Round	December 18, 2026

STAGES & PROCEDURES

1. Registration Flow

The following are the registration steps for the Automation Challenge COLLEGE – Microcontroller Competition IARC XII:

a. Register through the following link: https://its.id/m/FormRegistIoTIARCXII

b. Participants register online by filling out the provided form
c. Pay the registration fee

CATEGORY	BATCH	REGISTRATION FEE
COLLEGE	1	IDR. 165.000
	2	IDR. 195.000
	3	IDR. 235.000

d. Further information will be sent via personal email and the official IARC website.

2. Preliminary Round

The following are the stages of the preliminary round for the Automation Challenge Advance – Microcontroller Competition IARC XII:

a. During the online preliminary stage, participants will be provided with several case studies and must choose one of them to be solved using a microcontroller program.
b. Participants are required to draft a proposal following the structure outlined in the Proposal Structure section and formatted according to the Proposal Format Guidelines.
c. Participants must submit their completed proposals in accordance with the competition rules.
d. Participants are required to submit a video explaining the design presented in the proposal.

e. The 10 teams with the highest cumulative scores from the proposal and video will advance to the final round.
f. All updates and official announcements will be shared via the official IARC Instagram account (@iarc_its).

3. Final Round

The following are the stages of the final round for the Automation Challenge Advance – Microcontroller Competition IARC XII:

a. Teams that qualify for the final round are required to attend the Technical Meeting on December 18, 2026, for more detailed information regarding the final round.
b. Finalists will be given time to realize the proposal submitted during the preliminary round into a physical prototype before showcasing it on the day of the final round.
c. Each team must bring an X-Banner displaying information related to their prototype to their booth.
d. The criteria and weighting for the final round evaluation will be announced during the Technical Meeting on December 18, 2026.

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COMPETITION TECHNICAL

A. **Participant Requirements**

a. Competition Participants

1. Participants must be active Diploma or Undergraduate (Bachelor) students from public or private universities across Indonesia, verified by a Student Identity Card (KTM) or an official active student certificate.
2. There is no limit to the number of teams that can be sent from each institution or university.
3. Each team consists of a maximum of three members, and all members must be from the same institution or university.
4. Each team must be accompanied by one faculty advisor. If an institution sends multiple teams, a single faculty advisor may supervise more than one team.

B. Project Submission Requirements

a. The idea or proposal submitted must be a microcontroller-based automation system design that corresponds to the advanced study case provided by the committee for the Undergraduate category (Automation Challenge Advance – Microcontroller Competition).

b. Each team is required to select and include one study case aligned with the overarching theme, *"Innovative Microcontroller for Sustainable Living: Energy, Disaster Mitigation, or Appropriate Technology,"* that is most relevant to their problem-solving approach.

c. The ideas or proposals submitted for the preliminary round must be in the form of a proposal and must strictly follow the format prescribed by the committee (see the Proposal Format Guidelines chapter).

d. Proposals that advance to the final round must be realized into a physical prototype or product demonstration and presented offline at the exhibition booth.

C. Administrative Rules

a. Every team must register in accordance with the timeline established by the committee.

b. Participants are required to develop and submit a proposal based on the advanced microcontroller study case, following the format and provisions set by the committee.
c. Participants must create a video explaining the microcontroller product they have developed.
d. Both the proposal and the video must be submitted via the following link: https://its.id/m/PENGUMPULANBABAKPENYISIHANAUTOMATIONCHALLENGE
e. Participants who qualify for the final round are required to attend the Technical Meeting on December 18, 2026.

TERMS AND CONDITIONS

1. Proposal Format and Requirements

A. The video and proposal must be created in proper and correct Indonesian, in accordance with the General Guidelines for Indonesian Spelling (PUEBI).

B. The proposal serves as primary evidence of the authenticity and feasibility of the team's work, as well as the basis for the committee's decision regarding advancement to the next stage.

C. The proposal will serve as the primary reference during the final round (presentation and product demonstration).

D. Proposal Structure Guidelines

1. Cover
2. Table of Contents
3. Introduction
   - Background (including the rationale for selecting the study case)
   - Proposed Idea/Concept
4. Problem Identification
5. Automation System & Microcontroller Methodology

   In this section, teams are required to provide a strong scientific and technical foundation for their proposed idea/concept. Teams must present a summary and synthesis of previous research, tool development, or literature relevant to the chosen study case topic.
6. Proposal Description (Interconnective System Analysis and Design)
7. Technical Details of the Prototype / Microcontroller Idea Implementation
8. Impact & Sustainability
9. Budget Plan

   In this section, teams are required to create a budget plan table using the following format: (**No.**, **Component**, **Price**, **Purchase Reference Link**).

NO	Component	Price	Reference link
1	ESP 32	IDR 100,000	LINK ESP32

The total overall budget for prototype components must not exceed IDR 500,000 (five hundred thousand rupiahs). This budget is strictly for prototype components and excludes shipping costs, X-Banners, or other supporting equipment.

10. Conclusion
11. References

E. Formatting Guidelines

1. Paper Size : A4
2. Font : Times New Roman, 12 pt (body text)
3. Line Spacing : 1.5
4. Margins :

   Left : 4 cm

   Right : 3 cm

   Top : 3 cm

   Bottom: 3 cm

5. Page Numbering: Starts from the Introduction section.

2. Video Format and Guidelines

A. Video Content

1. Team Introduction: Team name and institution/university origin.
2. Selected Subtheme: Energy, Disaster Mitigation, or Appropriate Technology, along with the proposed automation system innovation/design.

3. Explanation of Problem, Solution, and Advantages: Overview of the problem addressed, the solution provided, and the key advantages of the prototype described in the proposal.
4. Alignment with the Theme: The correlation and relevance of the innovation to the selected theme.

B. Video Guidelines

1. Videos must be created as engagingly as possible.
2. The maximum video duration is 10 minutes.
3. Videos must be submitted by uploading them to YouTube.
4. The video aspect ratio must be 16:9.

XII IARC Industrial Automation and Robotic Competition

STUDY CASES

1. **Automatic Gas Leak Protection & Detection System** *(Subtheme: Energy and Mineral Resources / Energy)*

   The use of LPG or biogas in commercial kitchens or industrial processing facilities often presents the risks of fuel wastage and explosion hazards due to undetected pipeline leaks. Facility managers require a solution that not only provides early warnings but also acts swiftly to shut off the gas supply before energy loss and fire hazards occur. Design a microcontroller-based gas safety and efficiency system that detects methane/LPG gas concentrations and abnormal room temperature increases using sensors. The system must automatically trigger actuators (a motorized valve) to shut off the main gas pipeline valve and activate an exhaust fan. Furthermore, the system must allow managers to monitor gas concentrations in real-time and receive hazard notifications via a mobile/laptop dashboard.

2. **Weather-Affected Agricultural Soil pH & Moisture Monitoring System** *(Subtheme: Disaster Mitigation and Climate Change)*

   Climate change and prolonged droughts cause agricultural soil to lose moisture rapidly and experience shifts in acidity levels (pH), leading to crop failure and barren soil. Farmers require a practical soil condition monitoring system capable of automatically balancing water content and soil chemistry. Design a microcontroller-based soil conditioning system that measures soil pH and moisture using appropriate sensors. If the soil is detected to be overly dry or acidic, the system must automatically trigger actuators (such as relays and mini pumps) to spray water or a neutralizing solution until optimal soil conditions are restored. Additionally, farmers must be able to monitor real-time soil condition data and watering history through a mobile or laptop dashboard.

3. **Automatic Multi-Component Animal Feed Formulation & Dosing System** *(Subtheme: Appropriate Technology)*

   Medium-scale livestock farmers frequently face inconsistencies in feed nutrient ratios due to manual mixing processes of raw materials (such as ground corn, rice bran, and concentrates) based on rough estimates. Business owners require a precision dosing and mixing device to ensure consistent feed composition and cost efficiency. Design a microcontroller-based feed processing system that detects the flow rate and cumulative weight of each component using load cell sensors on storage hoppers. The system must control screw feeder motors proportionally to dispense each raw material according to the target formula, followed by activating a mixing motor for a measured duration. Furthermore, business owners must be able to monitor the mixing ratio, total

materials used, and update formulation recipes in real-time via a mobile or laptop dashboard.

4. **Embedded AI and Smart Power Management for Industrial Equipment** *(Subtheme: Energy and Mineral Resources)*

   Production facilities frequently experience electricity cost spikes due to inefficient power usage in large-scale industrial loads that are left running during idle states. Conventional automatic power-cut systems using simple timers often disrupt ongoing work processes. Companies require a microcontroller-based control module equipped with AI-driven power consumption pattern prediction algorithms. Design a microcontroller-based smart load control system that analyzes current, voltage, and equipment usage trends in real-time. The system must utilize predictive algorithms to autonomously determine when to activate energy-saving modes (sleep mode) or redistribute loads without interrupting active operational workflows. Additionally, facility operators must be able to monitor power consumption analytics and system status in real-time via a mobile/laptop dashboard.

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PRELIMINARY ROUND ASSESSMENT

Assesment Criteria	Assesment Weights	Description
Inovation	30%	The level of novelty of the idea, creativity of the solution, and originality of the proposal.
Relation to the theme	20%	The alignment of the proposal with the "Innovative Microcontroller for Sustainable Living: Energy, Disaster Mitigation, and Appropriate Technology" objective, and its tangible contribution to addressing relevant local or national issues.
Impact & Sustainability	20%	The proposal's potential benefits for society, the environment, or industry, as well as opportunities for long-term sustainability.
Creative Video	20%	Ability to explain ideas and their connection to the chosen theme.
Format compliance	10%	Adherence to writing procedures and formatting, the structural organization of the proposal, and the use of correct language and spelling in accordance with the guidelines.